

**ICCS121 Sys Skills & Low-lvl Prog
Quiz 1**

Name (Print): _____
Student ID: _____

Directions:

- This exam contains 7 pages (including this cover page) and 4 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.
- No collaboration of any kind is allowed.
- To earn full credit on open-ended questions, you must show sufficient work to justify your answers.
- You have total of 120 minutes to complete the exam. You may submit your exam anytime within the time limit. Good luck!

Problem	Points	Score
1	14	
2	14	
3	11	
4	11	
Total:	50	

1. (14 points) Briefly explain what each of the following commands does.

(a) `cd ~`

(b) `mkdir temp`

(c) `ls -al .`

(d) `cp ../a.txt .`

(e) `mv a b`

(f) `ssh admin@10.1.2.1`

(g) `echo $PATH`

(h) `echo "Hello World" >> welcome.txt`

(i) `cat a.txt b.txt > c.txt`

(j) `cat hello.txt | grep "Hi"`

(k) `chmod 755 test.sh`

(l) `rm -rf ../home`

(m) `touch me`

(n) `man man`

2. (14 points) Write a bash script `preview.sh` to print out first line of all `.txt` files in a given directory. Suppose a directory called `/tmp/q1` has 4 files: `a.txt` `b.txt` `c.txt` and `d.dat` and the files are generated by the following commands:

```
echo "A\nA\nA\n" > a.txt
echo "B\nB\nB\n" > b.txt
echo "C\nC\nC\n" > c.txt
echo "D\nD\nD\n" > d.dat
```

We expect the following output.

```
./preview.sh /tmp/q1
A
B
C
```

Here are some useful information:

To list all files with extension `csv` in the current working directory:

```
for i in *.csv;
do
    echo $i
done
```

The following command prints out first 10 lines of the file `b1.csv`:

```
head -10 b1.csv
```

Don't forget to put a shebang at the top of your script.

3. (11 points) Answer the following questions

(a) Give a string that matches with the following regex.

`x.+y`

(b) Give a string that matches with the following regex.

`a[1-3].{2}c`

(c) Give a string that matches with the following regex.

`[^0-9]+_[a-z]{4,5}`

(d) Give a regex that matches valid MUIC student IDs. A valid MUIC student ID starts with 'u' and the third digit is always 8 e.g. u6881923, u5781234, u5581202.

(e) Give a regex that matches binary numbers i.e. strings containing only 0 and 1.

4. (11 points) Basic Git commands

(a) What is the command to initialize a git repository?

(b) What is the command to sync local commits to a remote repository?

(c) What is the command to create a local commit?

(d) What is the command to make a copy of a remote repository on your local machine?

(e) What is the command to include a changed file in the working directory to the staging?

Anchors		Quantifiers		Groups and Ranges	
^	Start of string	*	0 or more	.	Any character except new line (\n)
\A	Start of string	+	1 or more	(a b)	a or b
\$	End of string	?	0 or 1	(...)	Group
\Z	End of string	{3}	Exactly 3	(?:...)	Passive Group
\b	Word boundary	{3,}	3 or more	[abc]	Range (a or b or c)
\B	Not word boundary	{3,5}	3, 4 or 5	[^abc]	Not a or b or c
\<	Start of word			[a-q]	Letter between a and q
\>	End of word			[A-Q]	Upper case letter between A and Q
		Quantifier Modifiers		[0-7]	Digit between 0 and 7
Character Classes		"x" below represents a quantifier		\n	nth group/subpattern
\c	Control character	x?	Ungreedy version of "x"		Note: Ranges are inclusive.
\s	White space			Pattern Modifiers	
\S	Not white space	Escape Character		g	Global match
\d	Digit	\	Escape Character	i	Case-insensitive
\D	Not digit			m	Multiple lines
\w	Word	Metacharacters (must be escaped)		s	Treat string as single line
\W	Not word	^	[	.	
\x	Hexadecimal digit	\$	{	*	
\O	Octal digit	(	\	+	
		)		?	
POSIX		<	>		
[:upper:]	Upper case letters			e	Evaluate replacement
[:lower:]	Lower case letters			U	Ungreedy pattern
[:alpha:]	All letters	Special Characters		String Replacement (Backreferences)	
[:alnum:]	Digits and letters	\n	New line	\$n	nth non-passive group
[:digit:]	Digits	\r	Carriage return	\$2	"xyz" in /^(abc(xyz))\$/
[:xdigit:]	Hexadecimal digits	\t	Tab	\$1	"xyz" in /^(?:abc)(xyz)\$/
[:punct:]	Punctuation	\v	Vertical tab	\$`	Before matched string
[:blank:]	Space and tab	\f	Form feed	\$'	After matched string
[:space:]	Blank characters	\xxx	Octal character xxx	\$+	Last matched string
[:cntrl:]	Control characters	\xhh	Hex character hh	\$&	Entire matched string
[:graph:]	Printed characters				
[:print:]	Printed characters and spaces	Sample Patterns			
[:word:]	Digits, letters and underscore	Pattern	Will Match		
Assertions		([A-Za-z0-9-]+)	Letters, numbers and hyphens		
?=	Lookahead assertion	(\d{1,2}\V\d{1,2}\V\d{4})	Date (e.g. 21/3/2006)		
?!	Negative lookahead	([^\s]+(?:=\.(jpg gif png))\.\2)	jpg, gif or png image		
?<=	Lookbehind assertion	(^[1-9]{1}\$ ^[1-4]{1}[0-9]{1}\$ ^[50]\$)	Any number from 1 to 50 inclusive		
?!= or ?<!	Negative lookbehind	(#?([A-Fa-f0-9]){3}((([A-Fa-f0-9]){3})?))	Valid hexadecimal colour code		
?>	Once-only Subexpression	((?=[*\d])(?=[*a-z])(?=[*A-Z]).{8,15})	String with at least one upper case letter, one lower case letter, and one digit (useful for passwords).		
?()	Condition [if then]	(\w+@[a-zA-Z_]+?\.[a-zA-Z]{2,6})	Email addresses		
?()	Condition [if then else]	(\<(/?[^\>]+\>)	HTML Tags		
?#	Comment				
		Note: These patterns are intended for reference purposes and have not been extensively tested. Please use with caution and test thoroughly before use.			